

Concise Writing



delete

remove, abrogate, abrogate, edit, out, blue, pencil, uproot, nullify, rescission, erase, decimate, expunge, obliterate, expulsion, override, revocable, recantation, cancel, annul, cull, edit, abolish, repeal, cut out, efface, cut out, eliminate, riddance, exterminate, extirpate, rescind, excise, revoke, annihilate



CONCISION

Being concise is essential in scientific writing. It helps your findings stand out and also enables the reader to find and absorb information quickly. There are many ways to revise your writing and make it more concise:

- **Deleting** words, phrases and sentences, but without making any other further changes.
- **Replacing** a phrase with a single word, again without making other changes.
- **Replacing** nouns with verbs, or verbs with nouns

Concision in Individual sentences

Replacing a noun phrase with a verb or can

1. The system **is installed** automatically.

1. **The installation of the system is done** automatically.

2. These systems **can be managed by**.

2. **The management of these systems can be done** by ...

3. This section **explains** the various parameters.

3. This section **contains an explanation of** the various parameters.

4. These methods will be used **to investigate** the properties of

4. These methods will be used **for an investigation of** the properties of...

5. These are used as markers, so that they **can be detected** at low levels.

5. These are used as markers, **thus making possible their detection** at low levels.

6. The aim of this document is **to evaluate** new solutions for.

6. The aim of this document is **the evaluation of** new solutions for ...

7. Note that the user **can insert** a new value .

7. Note that **the insertion of a new value can be performed by the user** .

8. I will outline the essential characteristics **of application, registration and protection of**

I will outline the essential characteristics **for applying, registering and protecting** a trademark

Replacing a noun phrase with a verb or *can*

E.g. A comparison of x and y *was made*. = x and y were compared.

1. The user **can** resend the email again

1. The user **has the ability to** re-send the email a

2. **The automatic pricing is configured** using the xyz file.

2. **The configuration of the automatic pricing is done** using the xyz file.

3. This takes place **while connections** between X and Y **are being set up**

3. This takes place **during the setup of connections** between X and Y.

4. The first step is **the choice of** X and Y.

4. The first step is **to choose** X and Y.

5. This is determined by **the choice of** X.

5. This is determined **by choosing** X.

6. This highlights **a much better performance of X** compa

6. This highlights **that X performed better than**

7. **After the set up of the system** , X is ...

7. **After the system has been set up** , X is

8. Particular care has been taken **in the design of X**.

8. Particular care has been taken **in designing X** .

9. X **presents customisability features**.

9. X **can be customized**.

Replacing several words with one word

1. This one was bigger **with respect to** the other one.

1. This one was bigger **than** the other one.

2. This has **made it possible for** us to do ...

2. This has **enabled** us to do ...

3. It is possible **on the condition that** the cost is specific

3. It is possible **provided that** the cost is specified as well.

4. We **found it useful to consider** the three cases separately.

4. We **considered** the three cases separately.

5. The time interval between **the time instant**

5. The time interval between **when** a packet arrives and **when** it leaves

6. **From now on** these will be referred to as X and Y.

6. **Hereafter** these will be referred to as X and Y.

7. This **allows us to reduce** the complexity.

7. This **reduces** the complexity.

8. This is an interesting theory and is similar to **the one p**

8. This is an interesting theory and is similar to **Smith's**.

9. This was done **by means of** the tracking feature.

9. This was done **using / with** the tracking feature.

10. Recently **the use of** such methods has led to

10. Recently **using** such methods has led to unexpected side-effect

11. This happened **owing to the fact that** the samples

11. This happened **because** the samples were not uniform.

12. **It should furthermore be underlined** that the samples were not

12. **Note** that the samples were not uniform.

13. There are two journals explicitly

13. There are two journals explicitly devoted to **(studying)** branding strategies.

14. This number **was equal to** 47% in low-income countries.

14. This number **was** 47% in low-income countries.

Deleting unnecessary words and phrases. Delete any possible excess language in these sentences then check the answers.

1. As we have already noted in Section 4.2.1, the presence of x can influence y.
2. However, we have to make use of other techniques.
3. Paint samples, as described previously, normally contain mixtures of different substances.
4. In comparative terms, there is no real difference between x and y.
5. Also, we present simulation results that will provide a twofold contribution: (1) confirm ..
6. The document was written in the English language and the contents represent anew innovation in the sector of telecommunications.
7. [In our documents,] we only use the Track Changes functionality for revisions.
8. Rows and also cells are highlighted in different colors in order to give a more effective view.
9. We identified a number of key factors that might affect, at least in principle, the performance.
10. They have shown [that] we should also consider the possibility of doing the testing in advance.
11. It is important to note that one plus one is equal to two.
12. In general, the phenomenon of e-commerce is often characterized by the absence of a direct contact between seller and buyer [Mugandi, 2016].
13. In particular, the scope of this section is to show the strengths of ...
14. Most of the organic materials that can be encountered in a painting are characterised by a macromolecular nature.
15. The use of this method is recommended in all cases where $x = y$.

1. ~~As we have already~~ noted in Section 4.2.1, the presence of x can influence y.
2. However, we have to ~~make~~ use of other techniques.
3. Paint samples, ~~as described previously,~~ normally contain mixtures of different substances.
4. ~~In comparative terms,~~ there is no real difference between x and y.
5. Also, we present simulation results that ~~will provide a twofold contribution:~~ (1) confirm ..
6. The document was written in ~~the~~ English ~~language~~ and the contents represent a new innovation in ~~the sector of~~ telecommunications.
7. [In our documents,] we only use ~~the~~ Track Changes ~~functionality~~ for revisions.
8. Rows and ~~also~~ cells are highlighted in different colors ~~in order~~ to give a more effective view.
9. We identified ~~a number of~~ key factors that might affect, ~~at least in principle,~~ the performance.
10. They have shown [that] we should ~~also~~ consider ~~the possibility of doing the~~ testing in advance.
11. ~~It is important to~~ note that one plus one is equal to two.
12. ~~In general, the phenomenon of~~ e-commerce is often characterized by the absence of a direct contact between seller and buyer [Mugandi, 2016].
13. ~~In particular,~~ the scope of this section is to show the strengths of ...
14. Most of ~~the~~ organic materials ~~that can be~~ encountered in a painting are characterised by a macromolecular nature.
15. ~~The use of~~ this method is recommended in all cases where $x=y$.

Removing redundant words when referring to the literature

~~It has been shown that~~ microwaves significantly shorten reaction times, giving good reaction yields and a reduced decomposition of labile sugars [14–16].

~~Smith and Jones proved that~~ x is greater than y . In particular, they demonstrated that

1. In this paper we give an overview of the most relevant existing work in this area of research.
2. It is in fact known that glucose is one of the products of photosynthesis in plants [Yang, 2014].
3. Experience teaches us that this is generally the best approach (see our previous papers: 12, 22, and 34).
4. Several authors in the past and also more recently have shown that substantial improvements in performance can ...
5. Several algorithms have been proposed during the last few years and their properties have been thoroughly analyzed in the literature [3].
6. It should be borne in mind that there may be a high environmental contamination level [Smrkolj, 2005].
7. Finally, it is known that in archaeological waxes, hydrocarbons sublime over time and it cannot be excluded that a similar phenomenon may occur in a thin film of oil [Bakali, 2014].
8. Another proposal is presented in [12], where the authors use a timeseries analysis with travel speed simulation to predict future trajectories. In particular, they use a process based on range querying with spatialtemporal constraints on moving object database.

1. In this paper we ~~give an~~ overview of the most relevant existing work in this area of research.
2. ~~It is in fact known that~~ glucose is one of the products of photosynthesis in plants [Yang, 2014].
3. ~~Experience teaches us that~~ this is generally the best approach (see our previous papers: 12, 22, and 34).
4. Several authors ~~in the past and also more recently~~ have shown that substantial improvements in performance can ...
5. Several algorithms have been proposed ~~during the last few years~~ and their properties have been thoroughly analyzed ~~in the literature~~ [3]. / Several algorithms have been proposed ~~during the last few years~~ and their properties have been thoroughly analyzed ~~in the literature~~ [3].
6. ~~It should be borne in mind that~~ there may be a high environmental contamination level [Smrkolj, 2005].
7. Finally, ~~it is known that~~ in archaeological waxes, hydrocarbons sublime over time and ~~it cannot be excluded that~~ a similar phenomenon may occur in a thin film of oil [Bakali, 2014].
8. ~~Another proposal is presented in~~ [12], where the authors use a time-series analysis with travel speed simulation to predict future trajectories. ~~In particular,~~ they use a process based on range querying with spatial-temporal constraints on moving object database.

Wordiness- try to avoid

Wordy	This finding makes it apparent that... (6 words)	Verb not noun
Concise	This finding demonstrates that... (4 words)	
Wordy	We come to the conclusion that... (6 words)	Verb not noun
More	We conclude that... (3 words)	
Wordy	In a paper published by Bonderic et al., experiments were described that led to similar results.	
Concise	Bonderic et al., reported similar results.	
Wordy	In our judgment, this finding clearly demonstrates. (7 words)	
Concise	This finding clearly demonstrates... (4 words)	
Wordy	Kamiyama et al. did an analysis...	Verb not noun
Concise	Kamiyama et al. analysed...	
Wordy	A synthesis was done by Martinez et al. ...	Verb not noun
Concise	Martinez et al. synthesized...	
Wordy	The results, as illustrated in Table 3, suggest that... (9 words)	
Concise	The results (Table 3) suggest that... (6 words)	

Wordy Phrase	Concise Alternative	Wordy Phrase	Concise Alternative
a number of	many, several	in order to	to
based on the fact that	because	in spite of the fact that	although
by means of	by	is/are known to be	is/are
despite the fact that	although	it is clear that	clearly
due to the fact that	because	reported in the literature	reported
if it is assumed that	if	subsequent to	after

Wordy phrase	Comment
It is worth mentioning at this point..	Just state the point ...
It is interesting to note that..	Just state your point...
It may be said that..	Just state your point...
As already stated..	Do you need to repeat? Why? (As noted –Fig 3- ...)
It was demonstrated that..	Wang (2018) demonstrated
It has been found that..	Ping (2020) found
It has long been known that..	Huang (1980) first synthesized....

Source + finding

Concision in Paragraphs



After doing one-way ANOVA analysis, fight performance showed no significant difference between different CDI types under the same N-Back task ($p>0.05$) (shown in Table 6). The mean flight performance score on the digital CDI was higher than that on the analogue one (as shown in Fig 41), which satisfied HPPF and indicated participants could fly 'better' with a digital CDI. In other words, the required resources under the digital CDI has not exceeded the maximum available resources but the spare capacity under the analogue one was close to zero (as shown in Fig 15). (94 words)

Unnecessary

Shorten

Word order
change

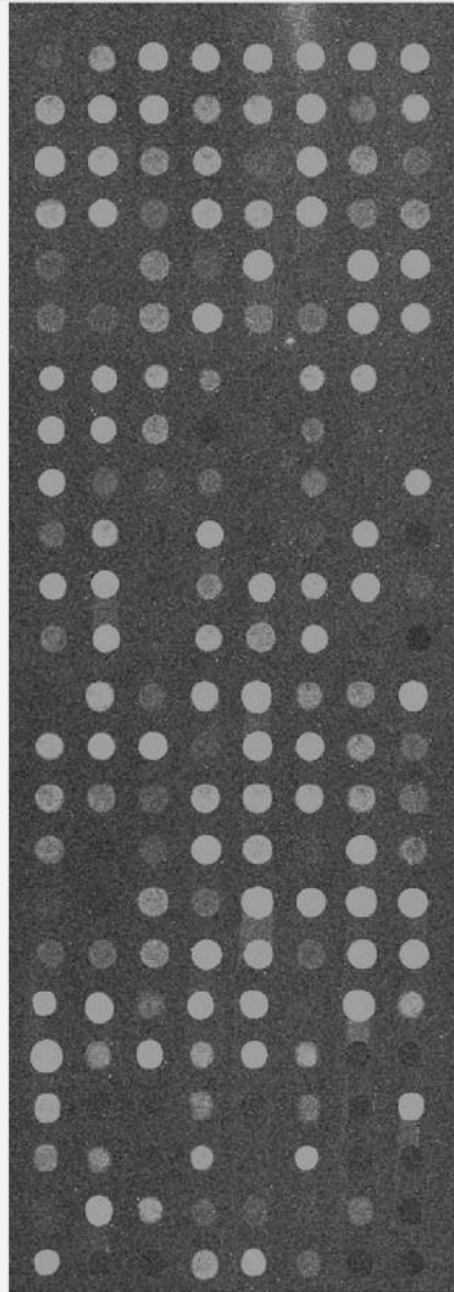
Step 1: ~~After doing~~ one-way ANOVA analysis showed no significant difference in fight performance between different CDI types under the same N-Back task ($p>0.05$) (~~shown in~~ Table 6). The mean flight performance score on the digital CDI was **higher** than that on the analogue one (~~as shown in~~ Fig 41), which satisfied HPPF and indicated participants could fly 'better' with a digital CDI. In other words, the required resources under the digital CDI did not exceed the maximum available resources but the spare capacity under the analogue condition was close to zero (~~as shown in~~ Fig 15).

Is this
correct?

Re-order to bring 'analogue'
and 'digital' together. Brings
verb earlier in sentence

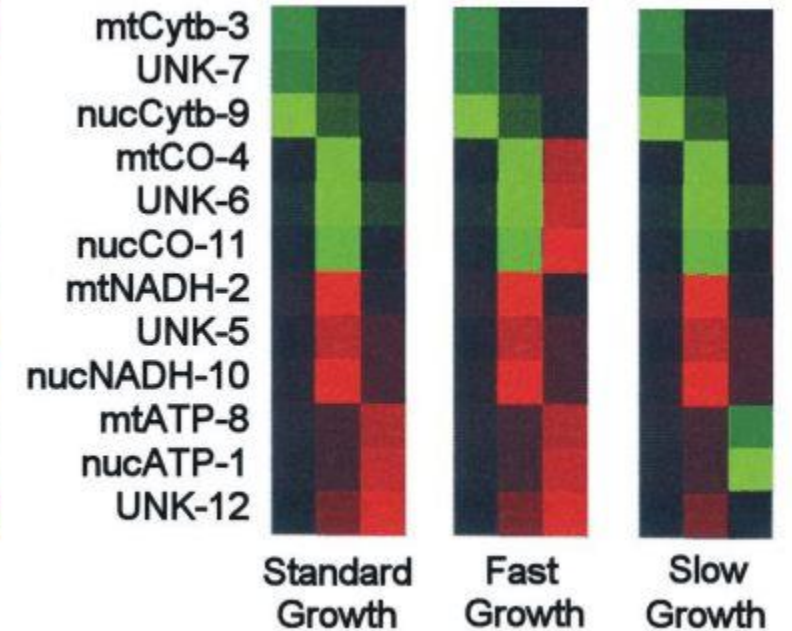
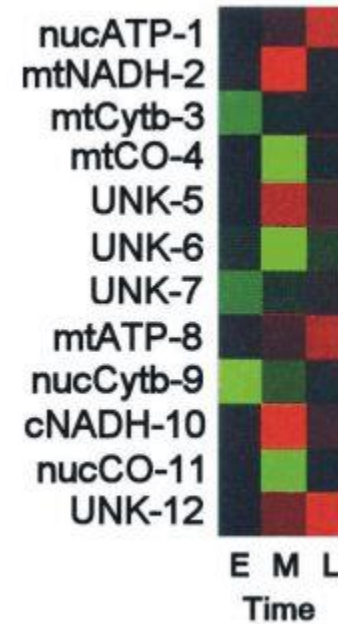
Step 2: One-way ANOVA analysis showed no significant difference in fight performance between CDI types under the same N-Back task ($p>0.05$) (Table 6). The mean flight performance score was higher on the digital than analogue CDI (Fig 41), which satisfied HPPF which indicated participants could fly 'better' with a digital CDI. **This means** the required resources under the digital CDI did not exceed the maximum available resources but the spare capacity under the analogue condition was close to zero (Fig 15). 79 words

Simple
past verb



m02	m14	m25	m32	m44	m53	m73	m86
m04	m16	m25.1	m34	m45.1	m55	m75	m88
m06	m18	m25.1	m36	m47	m57	m77	m90
m08	m20	m27	m38	m48.1	m59	m79	m92
m10	m22	m29	m40	m49	m69.1	m82	m94
m12	m23.1	m30	m42	m51	m71	m84	m96
m97	m113	m120	m131	m144	m156	m168	m145.4
m99	m115	m122	m134	m146	m158	m170	m34.2
m102	m117	m124	m136	m148	m160	m154.3	m166.5
m104	m118-9	m125	m138	m150	m162	m38.5	water
m106	m119.2	m127	m140	m152	m164	m84.2	Snx 2
m108	m119.4	m129	m142	m154	m166	m132.1	water
m01	m13	m24	m31	m43	m52	m72	m85
m03	m15	m25.1	m33	m45	m54	m74	m87
m05	m17	m25.2	m35	m46	m56	m76	m89
m07	m19	m26	m37	m48	m58	m78	m91
m09	m21	m28	m39	m48.2	m69	m80	m93
m11	m23	m29.1	m41	m50	m70	m83	m95
m98	m114	m121	m132-3	m145	m157	m169	m154.4
m100	m116	m123	m135	m147	m159	m26.1	m165.1
m103	m117.1	m124.1	m137	m149	m161	m36.1	rp10
m105	m119.1	m126	m139	m151	m163	m44.1	water
m107	m119.3	m128	m141	m153	m165	m120.1	water
m112	m119.5	m130	m143	m155	m167	m141.1	water

Gene analysis



Read the following extra from the methodology section of a student dissertation. Try to identify words and phrases that could be eliminated to make the writing more concise.

The available data sets do not have a common format and as a result, all collected data needs to be converted into a standard format for ML methods. We encountered many data consistency problems including sequencing, different experimental samples, different versions of gene sets and gene libraries across the studies, in addition to biological diversity and so on. Our group performed data standardization and converted all data sets into a unified feature list and comparable counts (also called the 'raw counts scale'). The standardized data was in the form of sparse matrix, having 20,608 rows representing features (genes'), and up to 11 standardized data format individual cells. As can be seen in figure 5, we created standardized data format. Most values in each matrix are zeros. We estimate that the raw counts represent approximately 5-30% of genes that are expressed in the cells. In this project, we focused on 10x technology since it is relatively cheap, it also has a strict biological protocol libraries which cover the entire genome, and finally it can screen up to 100,000 cells in a single experiment. (196 words)

We + active verb

Our + verb

Data standardization

The standardized data

We + active verb

standardized data format

We + active verb

Language we can cut out

The available data sets do not have a common format and as a result, all collected data needs to be converted into a standard format for ML methods. We encountered many data consistency problems including sequencing methods, experimental samples, different versions of genome and gene libraries used in the studies, in addition to biological diversity and so on. Our group performed data standardization and converted data sets into a unified format with an identical feature list and raw counts (also called the 'raw counts scale'). The standardized data was in the form of a matrix, having 30,698 rows representing features (referring to 'genes'), and up to 100,000 columns representing individual cells. As can be seen in figure 3 below we created standardized data format. Most values in each matrix are zeros. We estimate that the raw counts represent approximately 5-30% of genes that are expressed in the cells. In this project we focused on 10x technology since it is relatively cheap, it also has a strict biological protocol, it contains gene libraries which cover the entire genome, and finally it can screen up to 100,000 cells in a single experiment [3]. (196 words)

We can assume or infer the connection between the 2 sentences

It's obvious so no need to use a verb phrase

Never use this phrase.

Good use of active verb + subject

The reader can infer

The reader can infer

Simplify the reference

Simplify the list and use punctuation

B: More concise version

The available data sets do not have a common format. All collected data need to be converted into a standard format for ML methods. Data consistency problems **included** different sequencing methods, equipment, experimental samples, different versions of genome and gene libraries used across the studies, in addition to biological diversity. Our group performed data standardization and converted all data sets into a unified format with an identical and comparable counts (raw counts scale). The standardized data was in the form of a matrix, having 30,698 rows representing features (genes), and up to 11,739 columns representing individual cells. Figure 3 shows an example of a standardized data format. Most values in each matrix are zeros. We estimate that the raw counts represent approximately 5-20% of genes that are expressed in the cells. In this project, we focused on 10x technology since it is relatively cheap, has a strict biological protocol, gene libraries cover the entire genome, and it can screen up to 100,000 cells in a single experiment. (169 words)

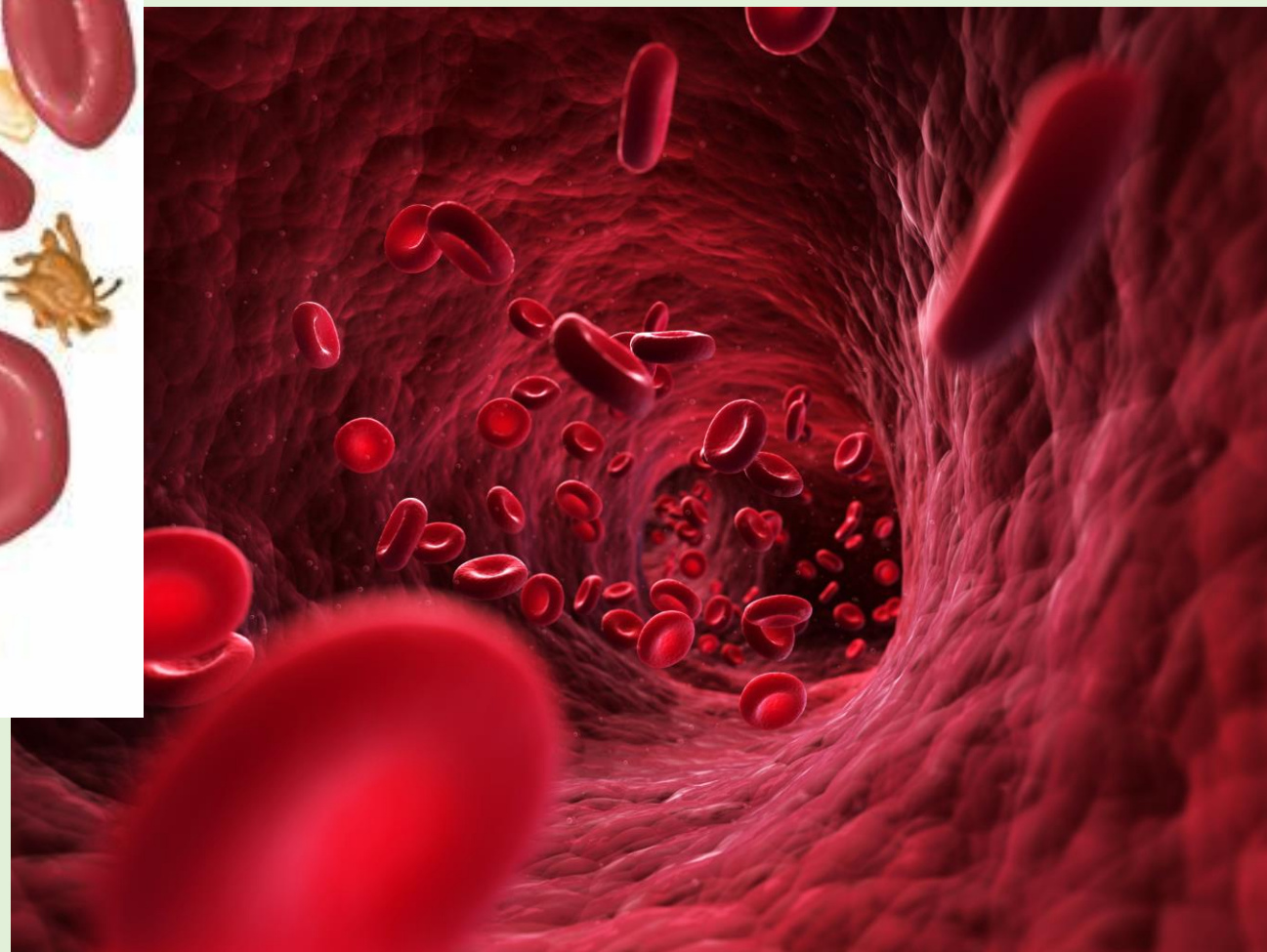
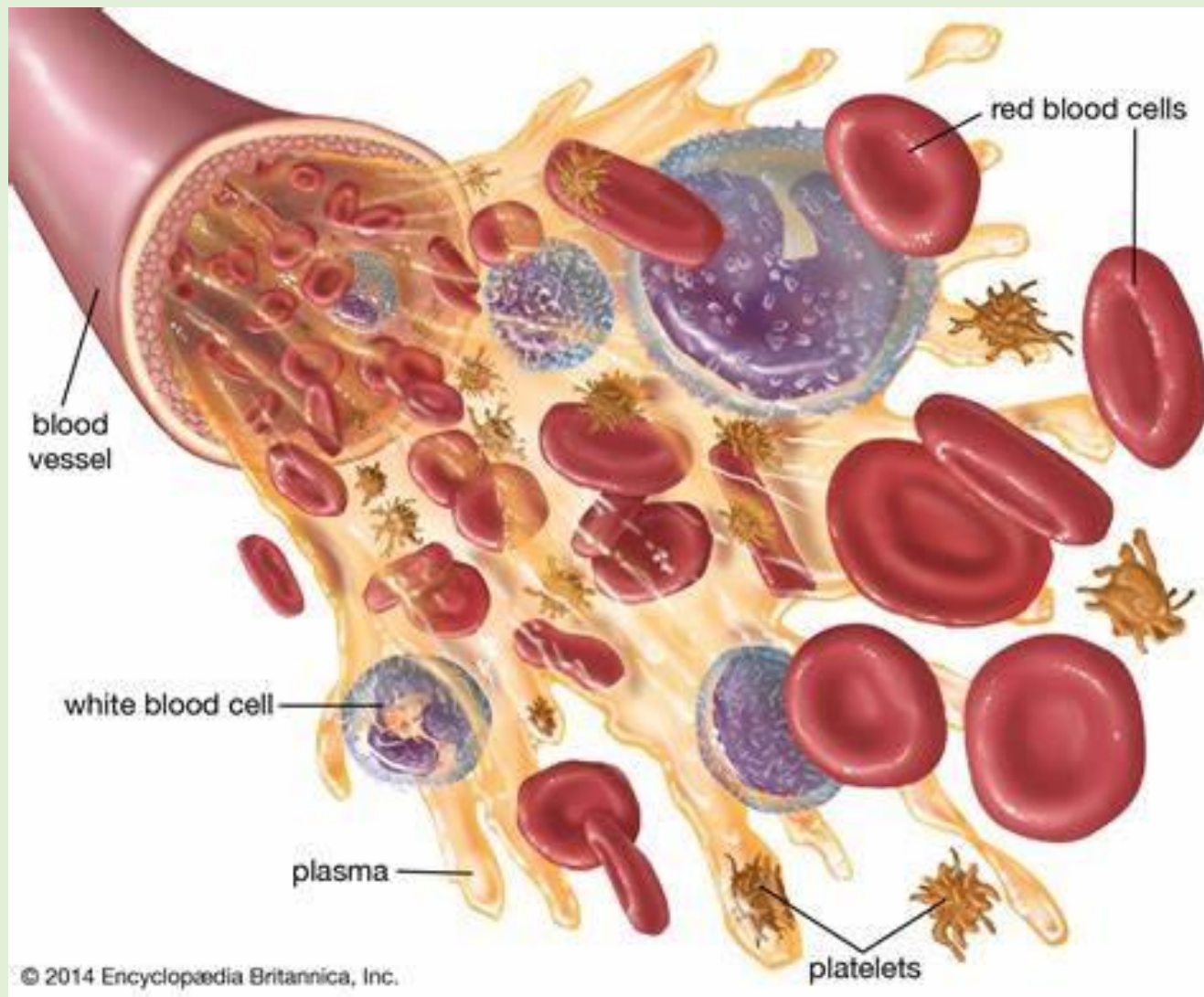
Introducing a list of items. Use a colon at the start of the list : and a semi-colon between the items of the list ;

Is this sentence necessary?

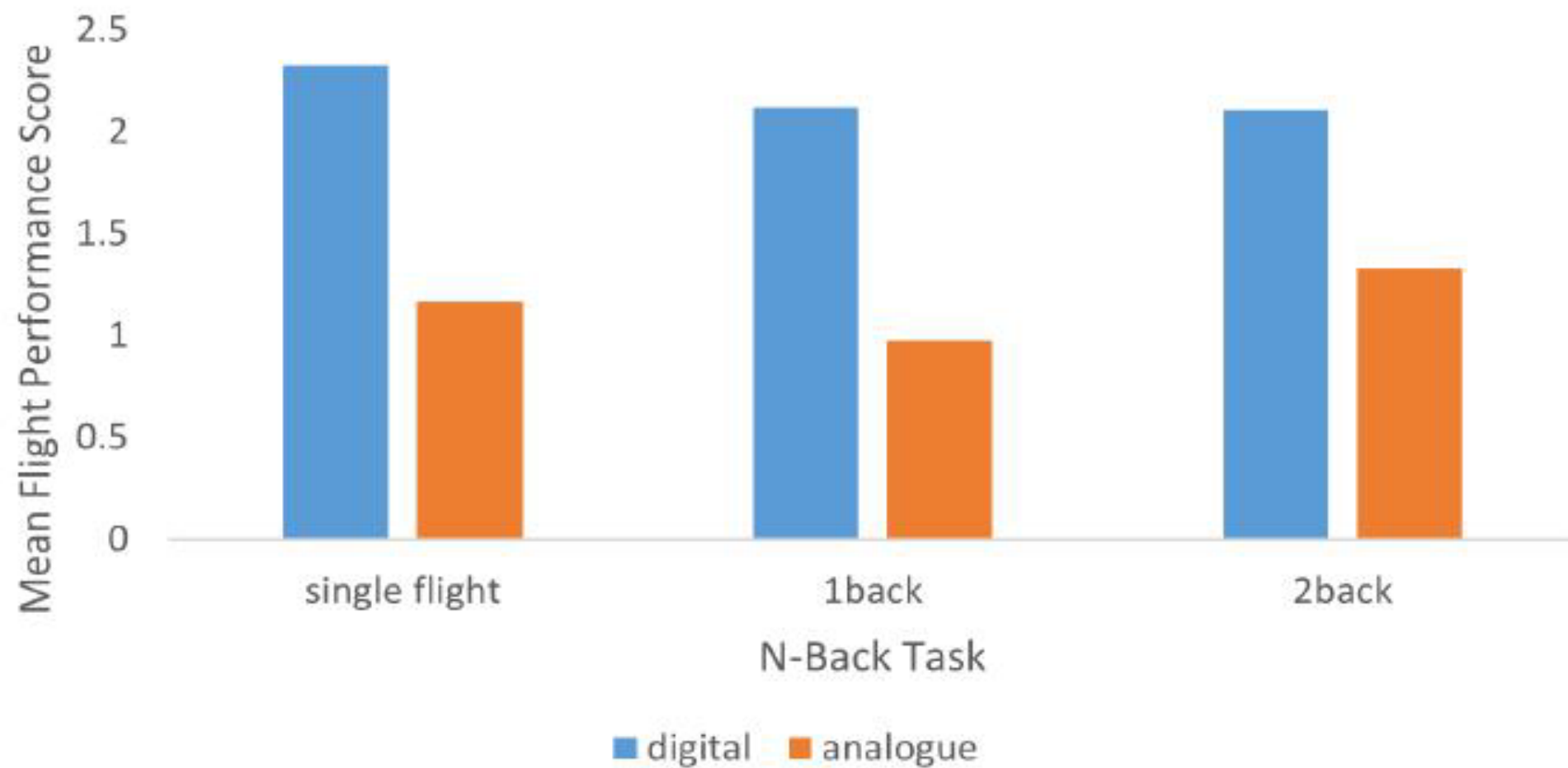
Overall, good, concise writing which cuts out unnecessary language-lean and mean!

Data consistency problems included: different sequencing methods; equipment; experimental samples; different versions of genome and gene libraries used across the studies; **and** biological diversity.

In this project, we focused on 10x technology since: it is relatively cheap; has a strict biological protocol; its gene libraries cover the entire genome; **and** it can screen up to 100,000 cells in a single experiment



Mean Score of the Flight Performance on different CDI



Read the following extra from the methodology section of a student dissertation. Try to identify words and phrases that could be eliminated to make the writing more concise.

Initially, data were extracted from three public SCT data sources, GEO database (GEOS data set) [34], Broad Institute database (BroadS1 and BroadS2 data sets) [35], and the 10x company demonstration data (10xS data set) [3]. After that, a total of 52 SCT data sets of PBMC were collected and prepared for the analysis which we carried out. These data sets were generated from fresh and frozen human blood samples using advanced 10x sequencing technology. Each data set has a metadata description. Metadata descriptions provide information about sample collection, processing, and the conditions of experiments. After we completed the metadata, we labeled each data set to reflect their PBMC cell types and subtypes, T cells(TC), B cells (BC), dendritic cells (DC), Monocytes (MC), and natural killer cells (NK). Finally, For classification, each data set was labeled with one of the five cell type labels. And For analysis, the cell subtype was also used for the assessment of correct classifications and misclassifications. (159)

La

No need to add 'stages' to this information- it's not an 'experiment'

Initially, data were extracted from three public SCT data sources, GEO (GEOS data set) [24], Broad Institute database (BroadS1 and BroadS2 data sets) [35], and the 10x Genomics data (10xS data set) [3]. **After that**, a total of 52 SCT data sets of PBMC were generated and prepared for the analysis **which we carried out**. These data sets were generated from fresh and frozen human blood samples using **advanced** 10x sequencing technology. Each data set has a **metadata description**. **Metadata descriptions** provide information about sample collection, processing, and the conditions of experiments. **After we completed the metadata**, we labeled each data set with PBMC cell types and subtypes, T cells(TC), B cells (BC), dendritic cells (DC), monocytes (MC), and natural killer cells (NK). **Finally**, for classification, each data set was labeled with one of the five labels. **And** for analysis, the cell subtype was also used for the assessment of correct classifications and misclassifications.

Superfluous- its obvious you 'did' something

Avoid unnecessary adjectives

No need to add 'stages'

No need to add 'stages'

Repetition of the same phrase. Need to rethink these 2 sentences

No need to add 'stages'

Avoid 'And' to start sentences. No linking word needed

B: More concise version

Data were extracted from three public SCT data sources, GEO database(GEOS data set) [34], Broad Institute database (BroadS1 and BroadS2 data sets) [35], and the 10x company demonstration data (10xS data set) [3]. A total of 52 SCT data sets of PBMC were collected and prepared for the analysis. These data sets were generated from fresh and frozen human blood samples using 10x sequencing technology. Each data set has a metadata description which provides information about sample collection, processing, and the conditions of experiments. We labeled each data set to reflect their PBMC cell types and subtypes, T cells(TC), B cells (BC), dendritic cells (DC), Monocytes (MC), and natural killer cells (NK). For classification, each data set was labeled with one of the five cell type labels. For analysis, the cell subtype was also used for the assessment of correct classifications and misclassifications. (143)

Overall, good, concise writing which cuts out unnecessary language

Editing your writing for concision

Referring to the literature

- How do you refer to figures and tables
- How do you refer to authors
- How do you use brackets

Subject choice

- Do you use 'We' 'Our' + an active verbs?

Nouns and Verbs

- Can you change some nouns to verbs to make the writing more 'active'

Redundant phrases

- Find and delete any phrases that are unnecessary- e.g. saying that X is 'relevant' or 'important' or 'noteworthy' etc.

Wordiness

- Find grammatical chunks and phrases that can be expressed in fewer words

Repetition of nouns or noun phrases

- Are any nouns or noun phrases repeated in adjacent sentences? Can you join the sentence to eliminate repetition?

